1) a)

$\int \sinh x \, \mathrm{d}x = \cosh x + c$

1) b)

$\int \cosh x \, \mathrm{d}x = \sinh x + c$

1) c)

$\int \operatorname{sech}^2 x \, \mathrm{d}x = \tanh x + c$

1) d)

$\int \sinh 3x \, \mathrm{d}x = \frac{1}{3}\cosh 3x + c$

1) e)

$\int \cosh \frac{x}{2} \mathrm{d}x = 2 \sinh \frac{x}{2} + c$

1) f)

$\int \operatorname{sech}^2 9x \, \mathrm{d}x = \frac{1}{9}\tanh 9x + c$

2)

$\int_{\ln 2}^{\ln 3} \sinh x \, \mathrm{d}x$

$= [\cosh x]_{\ln 2}^{\ln 3}$

$= (\cosh \ln 3) - (\cosh(\ln 2))$

$= \frac{e^{\ln 3} + e^{-\ln 3}}{2} - \frac{e^{\ln 2} + e^{-\ln 2}}{2}$

$= \frac{e^{\ln 3} + e^{\ln 3^{-1}}}{2} - \frac{e^{\ln 2} + e^{\ln 2^{-1}}}{2}$

$= \frac{e^{\ln 3} + e^{\ln \frac{1}{3}}}{2} - \frac{e^{\ln 2} + e^{\frac{1}{2}}}{2}$

$= \frac{3 + \frac{1}{3}}{2} - \frac{2 + \frac{1}{2}}{2}$

$= \frac{5}{3} - \frac{5}{4} = \frac{5}{12}$

3)

$$\int_{\ln 3}^{\ln 4} \cosh 2x \, \mathrm{d}x$$
$$= \left[\frac{1}{2}\sinh 2x\right]_{\ln 3}^{\ln 4}$$
$$= \left(\frac{1}{2}\sinh(2\ln 4)\right) - \left(\frac{1}{2}\sinh(2\ln 3)\right)$$
$$= \left(\frac{1}{2}\sinh(\ln 16)\right) - \left(\frac{1}{2}\sinh(\ln 9)\right)$$
$$= \frac{1}{2} \times \frac{e^{\ln 16} - e^{-\ln 16}}{2} - \frac{1}{2} \times \frac{e^{\ln 9} - e^{-\ln 9}}{2}$$
$$= \frac{e^{\ln 16} - e^{\ln 16^{-1}}}{4} - \frac{e^{\ln 9} - e^{\ln 9^{-1}}}{4}$$
$$= \frac{e^{\ln 16} - e^{\ln \frac{1}{16}}}{4} - \frac{e^{\ln 9} - e^{\ln \frac{1}{9}}}{4}$$
$$= \frac{16 - \frac{1}{16}}{4} - \frac{9 - \frac{1}{9}}{4}$$
$$= \frac{255}{64} - \frac{20}{9} = \frac{1015}{576}$$

4)

$$\int_0^{\ln 2} (15\cosh 3x - 9\sinh 3x) \, \mathrm{d}x$$
$$= [5\sinh 3x - 3\cosh 3x]_0^{\ln 2}$$
$$= (5\sinh(3\ln 2) - 3\cosh(3\ln 2)) - (5\sinh 0 - 3\cosh 0)$$
$$= (5\sinh(\ln 8) - 3\cosh(\ln 8)) - (0 - 3)$$
$$= 5 \times \frac{e^{\ln 8} - e^{-\ln 8}}{2} - 3 \times \frac{e^{\ln 8} + e^{-\ln 8}}{2} + 3$$
$$= 5 \times \frac{e^{\ln 8} - e^{\ln 8^{-1}}}{2} - 3 \times \frac{e^{\ln 8} + e^{\ln 8^{-1}}}{2} + 3$$
$$= 5 \times \frac{8 - \frac{1}{8}}{2} - 3 \times \frac{8 + \frac{1}{8}}{2} + 3$$
$$= \frac{315}{16} - \frac{195}{16} + 3$$
$$= \frac{21}{2}$$

5)

$\int_{-\ln 2}^{\ln 2} \operatorname{sech}^2 x \, \mathrm{d}x$

$= [\tanh x]_{-\ln 2}^{\ln 2}$

$= (\tanh(\ln 2)) - (\tanh(-\ln 2))$

$= \frac{e^{\ln 2} - e^{-\ln 2}}{e^{\ln 2} + e^{-\ln 2}} - \frac{e^{-\ln 2} - e^{-(-\ln 2)}}{e^{-\ln 2} + e^{-(-\ln 2)}}$

$= \frac{e^{\ln 2} - e^{\ln 2^{-1}}}{e^{\ln 2} + e^{\ln 2^{-1}}} - \frac{e^{\ln 2^{-1}} - e^{\ln 2}}{e^{\ln 2^{-1}} + e^{\ln 2}}$

$= \frac{e^{\ln 2} - e^{\ln \frac{1}{2}}}{e^{\ln 2} + e^{\ln \frac{1}{2}}} - \frac{e^{\ln \frac{1}{2}} - e^{\ln 2}}{e^{\ln \frac{1}{2}} + e^{\ln 2}}$

$= \frac{2 - \frac{1}{2}}{2 + \frac{1}{2}} - \frac{\frac{1}{2} - 2}{\frac{1}{2} + 2}$

$= \frac{3}{5} - \left(-\frac{3}{5}\right) = \frac{6}{5}$ as required

6)

$\lim_{\delta x \to 0} \sum_0^{\ln 2} (8 \cosh 4x - 5 \sinh 3x)$

$= \int_0^{\ln 2} (8 \cosh 4x - 5 \sinh 3x) \, \mathrm{d}x$

$= \left[2 \sinh 4x - \frac{5}{3} \cosh 3x\right]_0^{\ln 2}$

$= \left(2 \sinh(4 \ln 2) - \frac{5}{3} \cosh(3 \ln 2)\right) - \left(2 \sinh 0 - \frac{5}{3} \cosh 0\right)$

$= \left(2 \sinh(\ln 2^4) - \frac{5}{3} \cosh(\ln 2^3)\right) - \left(0 - \frac{5}{3}\right)$

$= \left(2 \sinh(\ln 16) - \frac{5}{3} \cosh(\ln 8)\right) + \frac{5}{3}$

$= 2 \times \frac{e^{\ln 16} - e^{-\ln 16}}{2} - \frac{5}{3} \times \frac{e^{\ln 8} + e^{-\ln 8}}{2} + \frac{5}{3}$

$= \left(e^{\ln 16} - e^{\ln 16^{-1}}\right) - \frac{5}{6}\left(e^{\ln 8} + e^{\ln \frac{1}{8}}\right) + \frac{5}{3}$

$= \left(16 - \frac{1}{16}\right) - \frac{5}{6}\left(8 + \frac{1}{8}\right) + \frac{5}{3}$

$= \frac{65}{6}$

7)

$\int_0^k (6\sinh 3x + 15\cosh 3x)\,\mathrm{d}x = 8$

$\Rightarrow [2\cosh 3x + 5\sinh 3x]_0^k = 8$

$\Rightarrow (2\cosh 3k + 5\sinh 3k) - (2\cosh 0 + 5\sinh 0) = 8$

$\Rightarrow 2 \times \frac{\mathrm{e}^{3k}+\mathrm{e}^{-3k}}{2} + 5 \times \frac{\mathrm{e}^{3k}-\mathrm{e}^{-3k}}{2} - 2 = 8$

$\Rightarrow 2\mathrm{e}^{3k} + 2\mathrm{e}^{-3k} + 5\mathrm{e}^{3k} - 5\mathrm{e}^{-3k} - 4 = 16$

$\Rightarrow 7\mathrm{e}^{3k} - 3\mathrm{e}^{-3k} - 20 = 0$

$\Rightarrow 7\mathrm{e}^{6k} - 3 - 20\mathrm{e}^{3k} = 0$

$\Rightarrow 7\mathrm{e}^{6k} - 20\mathrm{e}^{3k} - 3 = 0$

$\Rightarrow (7\mathrm{e}^{3k} + 1)(\mathrm{e}^{3k} - 3) = 0$

$\Rightarrow \mathrm{e}^{3k} = -\frac{1}{7}$ or $\mathrm{e}^{3k} = 3$

$\mathrm{e}^{3k} \neq -\frac{1}{7}$ as $\mathrm{e}^{3k} > 0$ for all real values of k

So $\mathrm{e}^{3k} = 3 \Rightarrow 3k = \ln 3$

$\Rightarrow k = \frac{1}{3}\ln 3$

8)

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} (\sec^2 x - \operatorname{sech}^2 x)\,\mathrm{d}x$$

$$= [\tan x - \tanh x]_{-\frac{\pi}{4}}^{\frac{\pi}{4}}$$

$$= \left(\tan\frac{\pi}{4} - \tanh\frac{\pi}{4}\right) - \left(\tan\left(-\frac{\pi}{4}\right) - \tanh\left(-\frac{\pi}{4}\right)\right)$$

$$= \left(1 - \frac{e^{\frac{\pi}{4}} - e^{-\frac{\pi}{4}}}{e^{\frac{\pi}{4}} + e^{-\frac{\pi}{4}}}\right) - \left(-1 - \frac{e^{-\frac{\pi}{4}} - e^{\frac{\pi}{4}}}{e^{-\frac{\pi}{4}} + e^{\frac{\pi}{4}}}\right)$$

$$= 2 + \frac{e^{-\frac{\pi}{4}} - e^{\frac{\pi}{4}}}{e^{\frac{\pi}{4}} + e^{-\frac{\pi}{4}}} - \frac{e^{\frac{\pi}{4}} - e^{-\frac{\pi}{4}}}{e^{\frac{\pi}{4}} + e^{-\frac{\pi}{4}}}$$

$$= 2 + \frac{e^{-\frac{\pi}{4}} - e^{\frac{\pi}{4}} - e^{\frac{\pi}{4}} + e^{-\frac{\pi}{4}}}{e^{\frac{\pi}{4}} + e^{-\frac{\pi}{4}}}$$

$$= 2 + \frac{2e^{-\frac{\pi}{4}} - 2e^{\frac{\pi}{4}}}{e^{\frac{\pi}{4}} + e^{-\frac{\pi}{4}}}$$

$$= \frac{2e^{\frac{\pi}{4}} + 2e^{-\frac{\pi}{4}}}{e^{\frac{\pi}{4}} + e^{-\frac{\pi}{4}}} + \frac{2e^{-\frac{\pi}{4}} - 2e^{\frac{\pi}{4}}}{e^{\frac{\pi}{4}} + e^{-\frac{\pi}{4}}}$$

$$= \frac{2e^{\frac{\pi}{4}} + 2e^{-\frac{\pi}{4}} + 2e^{-\frac{\pi}{4}} - 2e^{\frac{\pi}{4}}}{e^{\frac{\pi}{4}} + e^{-\frac{\pi}{4}}}$$

$$= \frac{4e^{-\frac{\pi}{4}}}{e^{\frac{\pi}{4}} + e^{-\frac{\pi}{4}}}$$

$$= \frac{4}{e^{\frac{\pi}{2}} + 1}$$

As $e^{\pi} + 1 > e^{\frac{\pi}{2}} + 1$, we must have that $\frac{4}{e^{\frac{\pi}{2}}+1} > \frac{4}{e^{\pi}+1}$

So $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} (\sec^2 x - \operatorname{sech}^2 x)\,\mathrm{d}x > \frac{4}{e^{\pi}+1}$ as required

9) a)

$$\int \sinh(2x + 5)\,\mathrm{d}x$$

$$= \frac{1}{2}\cosh(2x + 5) + c$$

9) b)

$$\int 8\cosh(3 - 4x)\,\mathrm{d}x$$

$$= -2\sinh(3 - 4x) + c$$

9) c)

$\int 2x \operatorname{sech}^2 x^2 \, dx$

$= \tanh x^2 + c$

9) d)

$\int x \sinh(3x^2 - 2) \, dx$

$= \frac{1}{6} \cosh(3x^2 - 2) + c$

9) e)

$\int \frac{\cosh \sqrt{x}}{2\sqrt{x}} dx$

$= \int \frac{1}{2} x^{-\frac{1}{2}} \cosh x^{\frac{1}{2}} \, dx$

$= \sinh \sqrt{x} + c$

9) f)

$\int \frac{\operatorname{sech}^2\left(\frac{3}{x^2}\right)}{4x^3} dx$

$= \int \frac{1}{4} x^{-3} \operatorname{sech}^2(3x^{-2}) \, dx$

$= \frac{1}{24} \tanh\left(\frac{3}{x^2}\right) + c$

10)

$$\int_0^{\ln 2} \sinh\left(\frac{3x+\ln 4}{2}\right) \mathrm{d}x$$
$$= \left[\frac{2}{3}\cosh\left(\frac{3x+\ln 4}{2}\right)\right]_0^{\ln 2}$$
$$= \left(\frac{2}{3}\cosh\left(\frac{3\ln 2+\ln 4}{2}\right)\right) - \left(\frac{2}{3}\cosh\left(\frac{0+\ln 4}{2}\right)\right)$$
$$= \left(\frac{2}{3}\cosh\left(\frac{\ln 8+\ln 4}{2}\right)\right) - \left(\frac{2}{3}\cosh\left(\frac{\ln 4}{2}\right)\right)$$
$$= \left(\frac{2}{3}\cosh\left(\frac{1}{2}\ln 32\right)\right) - \left(\frac{2}{3}\cosh\left(\frac{1}{2}\ln 4\right)\right)$$
$$= \left(\frac{2}{3}\cosh\left(\ln 32^{\frac{1}{2}}\right)\right) - \left(\frac{2}{3}\cosh\left(\ln 4^{\frac{1}{2}}\right)\right)$$
$$= \left(\frac{2}{3}\cosh\left(\ln\left(4\sqrt{2}\right)\right)\right) - \left(\frac{2}{3}\cosh(\ln 2)\right)$$
$$= \frac{2}{3}\times\frac{\mathrm{e}^{\ln\left(4\sqrt{2}\right)}+\mathrm{e}^{-\ln\left(4\sqrt{2}\right)}}{2} - \frac{2}{3}\times\frac{\mathrm{e}^{\ln 2}+\mathrm{e}^{-\ln 2}}{2}$$
$$= \frac{1}{3}\left(\mathrm{e}^{\ln\left(4\sqrt{2}\right)} + \mathrm{e}^{\ln\left(4\sqrt{2}\right)^{-1}}\right) - \frac{1}{3}\left(\mathrm{e}^{\ln 2} + \mathrm{e}^{\ln 2^{-1}}\right)$$
$$= \frac{1}{3}\left(\mathrm{e}^{\ln\left(4\sqrt{2}\right)} + \mathrm{e}^{\ln\left(\frac{1}{8}\sqrt{2}\right)}\right) - \frac{1}{3}\left(\mathrm{e}^{\ln 2} + \mathrm{e}^{\ln\frac{1}{2}}\right)$$
$$= \frac{1}{3}\left(4\sqrt{2} + \frac{1}{8}\sqrt{2}\right) - \frac{1}{3}\left(2 + \frac{1}{2}\right)$$
$$= \frac{11}{8}\sqrt{2} - \frac{5}{6}$$

11)

$$\int_0^{\sqrt{\ln 2}} 256x\cosh(4x^2)\,\mathrm{d}x$$
$$= \left[32\sinh(4x^2)\right]_0^{\sqrt{\ln 2}}$$
$$= \left(32\sinh\left(4\left(\sqrt{\ln 2}\right)^2\right)\right) - (32\sinh 0)$$
$$= (32\sinh(4\ln 2)) - 0$$
$$= 32\sinh(\ln 2^4)$$
$$= 32\sinh(\ln 16)$$
$$= 32\times\frac{\mathrm{e}^{\ln 16}-\mathrm{e}^{-\ln 16}}{2}$$
$$= 16\left(\mathrm{e}^{\ln 16} - \mathrm{e}^{\ln 16^{-1}}\right)$$
$$= 16\left(\mathrm{e}^{\ln 16} - \mathrm{e}^{\ln\frac{1}{16}}\right)$$
$$= 16\left(16 - \frac{1}{16}\right)$$
$$= 255$$

12)

$$\int_0^{\frac{\pi}{2}} \cos x \sinh(\sin x)\,\mathrm{d}x$$
$$= [\cosh(\sin x)]_0^{\frac{\pi}{2}}$$
$$= \left(\cosh\left(\sin\frac{\pi}{2}\right)\right) - (\cosh(\sin 0))$$
$$= \cosh 1 - \cosh 0$$
$$= \frac{e^1+e^{-1}}{2} - 1$$
$$= \frac{e^2+1}{2e} - \frac{2e}{2e}$$
$$= \frac{e^2-2e+1}{2e}$$
$$= \frac{(e-1)^2}{2e}$$

13) a)

$$\int \tanh x\,\mathrm{d}x$$
$$= \int \frac{\sinh x}{\cosh x}\mathrm{d}x$$
$$= \ln|\cosh x| + c$$

13) b)

$$\int \coth x\,\mathrm{d}x$$
$$= \int \frac{\cosh x}{\sinh x}\mathrm{d}x$$
$$= \ln|\sinh x| + c$$

14)

$\int_{\ln 2}^{\ln 4}(2\tanh 2x + 2\coth 2x)\,\mathrm{d}x$

$= [\ln|\cosh 2x| + \ln|\sinh 2x|]_{\ln 2}^{\ln 4}$

$= (\ln|\cosh(2\ln 4)| + \ln|\sinh(2\ln 4)|) - (\ln|\cosh(2\ln 2)| + \ln|\sinh(2\ln 2)|)$

$= (\ln|\cosh(\ln 16)| + \ln|\sinh(\ln 16)|) - (\ln|\cosh(\ln 4)| + \ln|\sinh(\ln 4)|)$

$= \left(\ln\left|\frac{e^{\ln 16}+e^{-\ln 16}}{2}\right| + \ln\left|\frac{e^{\ln 16}-e^{-\ln 16}}{2}\right|\right) - \left(\ln\left|\frac{e^{\ln 4}+e^{-\ln 4}}{2}\right| + \ln\left|\frac{e^{\ln 4}-e^{-\ln 4}}{2}\right|\right)$

$= \left(\ln\left|\frac{e^{\ln 16}+e^{\ln 16^{-1}}}{2}\right| + \ln\left|\frac{e^{\ln 16}-e^{\ln 16^{-1}}}{2}\right|\right) - \left(\ln\left|\frac{e^{\ln 4}+e^{\ln 4^{-1}}}{2}\right| + \ln\left|\frac{e^{\ln 4}-e^{\ln 4^{-1}}}{2}\right|\right)$

$= \left(\ln\left|\frac{16+\frac{1}{16}}{2}\right| + \ln\left|\frac{16-\frac{1}{16}}{2}\right|\right) - \left(\ln\left|\frac{4+\frac{1}{4}}{2}\right| + \ln\left|\frac{4-\frac{1}{4}}{2}\right|\right)$

$= \left(\ln\left(\frac{257}{32}\right) + \ln\left(\frac{255}{32}\right)\right) - \left(\ln\left(\frac{17}{8}\right) + \ln\left(\frac{15}{8}\right)\right)$

$= \ln\left(\frac{65535}{1024}\right) - \ln\left(\frac{255}{64}\right)$

$= \ln\left(\frac{257}{16}\right)$

So $p = 257$ and $q = 16$

15) a)

$\frac{\mathrm{d}}{\mathrm{d}x}(\operatorname{cosech} x) = \frac{\mathrm{d}}{\mathrm{d}x}((\sinh x)^{-1})$

$= -\cosh x\,(\sinh x)^{-2}$

$= -\frac{\cosh x}{\sinh^2 x}$

$= -\frac{1}{\sinh x} \times \frac{\cosh x}{\sinh x}$

$= -\operatorname{cosech} x \coth x$

15) b)

$\int \operatorname{cosech} x \coth x\,\mathrm{d}x = -\operatorname{cosech} x + c$

16) a)

$\frac{\mathrm{d}}{\mathrm{d}x}(\operatorname{sech} x) = \frac{\mathrm{d}}{\mathrm{d}x}((\cosh x)^{-1})$

$= -\sinh x\,(\cosh x)^{-2}$

$= -\frac{\sinh x}{\cosh^2 x}$

$= -\frac{1}{\cosh x} \times \frac{\sinh x}{\cosh x}$

$= -\operatorname{sech} x \tanh x$

16) b)

$\int \operatorname{sech} x \tanh x \, \mathrm{d}x = -\operatorname{sech} x + c$

17) a)

$\frac{\mathrm{d}}{\mathrm{d}x}(\coth x) = \frac{\mathrm{d}}{\mathrm{d}x}((\tanh x)^{-1})$

$= -\operatorname{sech}^2 x \, (\tanh x)^{-2}$

$= -\frac{\operatorname{sech}^2 x}{\tanh^2 x}$

$= -\frac{\frac{1}{\cosh^2 x}}{\frac{\sinh^2 x}{\cosh^2 x}}$

$= -\frac{1}{\sinh^2 x}$

$= -\operatorname{cosech}^2 x$

17) b)

$\int \operatorname{cosech}^2 x \, \mathrm{d}x = -\coth x + c$

18) a)

$\int x \sinh 3x \, \mathrm{d}x$

Use Integration by Parts:

$u = x$ $\qquad$ $\frac{\mathrm{d}v}{\mathrm{d}x} = \sinh 3x$

$\frac{\mathrm{d}u}{\mathrm{d}x} = 1$ $\qquad$ $v = \frac{1}{3}\cosh 3x$

$\therefore \int x \sinh 3x \, \mathrm{d}x = \frac{1}{3} x \cosh 3x - \int \frac{1}{3} \cosh 3x \, \mathrm{d}x$

$= \frac{1}{3} x \cosh 3x - \frac{1}{9} \sinh 3x + c$

18) b)

$\int x^2 \cosh\frac{x}{4}\,\mathrm{d}x$

Use Integration by Parts:

$u = x^2 \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \cosh\frac{x}{4}$

$\frac{\mathrm{d}u}{\mathrm{d}x} = 2x \qquad v = 4\sinh\frac{x}{4}$

$\therefore \int x^2 \cosh\frac{x}{4}\,\mathrm{d}x = 4x^2 \sinh\frac{x}{4} - \int 8x \sinh\frac{x}{4}\,\mathrm{d}x$

Use Integration by Parts again:

$u = 8x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \sinh\frac{x}{4}$

$\frac{\mathrm{d}u}{\mathrm{d}x} = 8 \qquad v = 4\cosh\frac{x}{4}$

$\therefore \int x^2 \cosh\frac{x}{4}\,\mathrm{d}x = 4x^2 \sinh\frac{x}{4} - \int 8x \sinh\frac{x}{4}\,\mathrm{d}x$

$= 4x^2 \sinh\frac{x}{4} - \left(32x \cosh\frac{x}{4} - \int 32\cosh\frac{x}{4}\,\mathrm{d}x\right)$

$= 4x^2 \sinh\frac{x}{4} - 32x\cosh\frac{x}{4} + \int 32\cosh\frac{x}{4}\,\mathrm{d}x$

$= 4x^2 \sinh\frac{x}{4} - 32x\cosh\frac{x}{4} + 128\sinh\frac{x}{4} + c$

19)

$\frac{\mathrm{d}y}{\mathrm{d}x} = 9(x+2)\sinh 3x$

Use Integration by Parts:

$u = 9(x+2) \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \sinh 3x$

$\frac{\mathrm{d}u}{\mathrm{d}x} = 9 \qquad v = \frac{1}{3}\cosh 3x$

$\therefore y = 3(x+2)\cosh 3x - \int 3\cosh 3x\,\mathrm{d}x$

$\Rightarrow y = 3(x+2)\cosh 3x - \sinh 3x + c$

The curve passes through $(0, 8)$:

$8 = 3(0+2)\cosh 0 - \sinh 0 + c$

$\Rightarrow 8 = 6 - 0 + c \Rightarrow c = 2$

$\therefore y = (3x+6)\cosh 3x - \sinh 3x + 2$

20) a)

$$\int e^x \sinh x \, dx$$
$$= \int e^x \times \frac{e^x - e^{-x}}{2} dx$$
$$= \frac{1}{2}\int e^x(e^x - e^{-x}) \, dx$$
$$= \frac{1}{2}\int (e^{2x} - 1) \, dx$$
$$= \frac{1}{4}e^{2x} - \frac{1}{2}x + c$$

20) b)

$$\int e^{2x} \cosh x \, dx$$
$$= \int e^{2x} \times \frac{e^x + e^{-x}}{2} dx$$
$$= \frac{1}{2}\int e^{2x}(e^x + e^{-x}) \, dx$$
$$= \frac{1}{2}\int (e^{3x} + e^x) \, dx$$
$$= \frac{1}{6}e^{3x} + \frac{1}{2}e^x + c$$

20) c)

$$\int e^{3x} \sinh 2x \, dx$$
$$= \int e^{3x} \times \frac{e^{2x} - e^{-2x}}{2} dx$$
$$= \frac{1}{2}\int e^{3x}(e^{2x} - e^{-2x}) \, dx$$
$$= \frac{1}{2}\int (e^{5x} - e^x) \, dx$$
$$= \frac{1}{10}e^{5x} - \frac{1}{2}e^x + c$$

20) d)

$$\int e^{-4x} \cosh 4x \, dx$$
$$= \int e^{-4x} \times \frac{e^{4x} + e^{-4x}}{2} dx$$
$$= \frac{1}{2}\int e^{-4x}(e^{4x} + e^{-4x}) \, dx$$
$$= \frac{1}{2}\int (1 + e^{-8x}) \, dx$$
$$= \frac{1}{2}x - \frac{1}{16}e^{-8x} + c$$

21)

$\cosh 2A \equiv 2\cosh^2 A - 1$

$\Rightarrow \cosh^2 A = \frac{1}{2}\cosh 2A + \frac{1}{2}$

$\therefore \int \cosh^2 x \, dx = \int \left(\frac{1}{2}\cosh 2x + \frac{1}{2}\right) dx$

$= \frac{1}{4}\sinh 2x + \frac{1}{2}x + c$

22)

$\cosh 2A \equiv 1 + 2\sinh^2 A$

$\Rightarrow \sinh^2 A = \frac{1}{2}\cosh 2A - \frac{1}{2}$

$\therefore \int \sinh^2 6x \, dx = \int \left(\frac{1}{2}\cosh 12x - \frac{1}{2}\right) dx$

$= \frac{1}{24}\sinh 12x - \frac{1}{2}x + c$

23)

$\cosh 2A \equiv 2\cosh^2 A - 1$

$\Rightarrow \cosh^2 A = \frac{1}{2}(\cosh 2A + 1)$

$\int_{\ln\frac{5}{6}}^{\ln\frac{6}{5}} \cosh^2 \frac{x}{2}\,\mathrm{d}x$

$= \int_{\ln\frac{5}{6}}^{\ln\frac{6}{5}} \left(\frac{1}{2}(\cosh x + 1)\right)\mathrm{d}x$

$= \frac{1}{2}\int_{\ln\frac{5}{6}}^{\ln\frac{6}{5}} (\cosh x + 1)\,\mathrm{d}x$

$= \frac{1}{2}\left[\sinh x + x\right]_{\ln\frac{5}{6}}^{\ln\frac{6}{5}}$

$= \frac{1}{2}\left(\left(\sinh\left(\ln\frac{6}{5}\right) + \ln\frac{6}{5}\right) - \left(\sinh\left(\ln\frac{5}{6}\right) + \ln\frac{5}{6}\right)\right)$

$= \frac{1}{2}\left(\left(\frac{\mathrm{e}^{\ln\frac{6}{5}} - \mathrm{e}^{-\ln\frac{6}{5}}}{2} + \ln\frac{6}{5}\right) - \left(\frac{\mathrm{e}^{\ln\frac{5}{6}} - \mathrm{e}^{-\ln\frac{5}{6}}}{2} + \ln\frac{5}{6}\right)\right)$

$= \frac{1}{2}\left(\left(\frac{\mathrm{e}^{\ln\frac{6}{5}} - \mathrm{e}^{\ln\frac{5}{6}}}{2} + \ln\frac{6}{5}\right) - \left(\frac{\mathrm{e}^{\ln\frac{5}{6}} - \mathrm{e}^{\ln\frac{6}{5}}}{2} + \ln\frac{5}{6}\right)\right)$

$= \frac{1}{2}\left(\left(\frac{\frac{6}{5} - \frac{5}{6}}{2} + \ln\frac{6}{5}\right) - \left(\frac{\frac{5}{6} - \frac{6}{5}}{2} + \ln\frac{5}{6}\right)\right)$

$= \frac{1}{2}\left(\left(\frac{11}{60} + \ln\frac{6}{5}\right) - \left(-\frac{11}{60} + \ln\frac{5}{6}\right)\right)$

$= \frac{1}{2}\left(\frac{11}{30} + \ln\frac{6}{5} - \ln\frac{5}{6}\right)$

$= \frac{11}{60} + \frac{1}{2}\ln\frac{36}{25}$

$= \frac{11}{60} + \ln\frac{6}{5}$

24)

$\cosh 2A \equiv 2\cosh^2 A - 1$

$\Rightarrow \cosh^2 A = \frac{1}{2}(\cosh 2A + 1)$

$\int_0^{\ln 3}(4\cosh^2 x - 3\cosh x)\,\mathrm{d}x$

$= \int_0^{\ln 3}\left(4 \times \frac{1}{2}(\cosh 2x + 1) - 3\cosh x\right)\mathrm{d}x$

$= \int_0^{\ln 3}(2\cosh 2x + 2 - 3\cosh x)\,\mathrm{d}x$

$= [\sinh 2x + 2x - 3\sinh x]_0^{\ln 3}$

$= (\sinh(2\ln 3) + 2\ln 3 - 3\sinh(\ln 3)) - (0 + 0 - 0)$

$= \frac{\mathrm{e}^{2\ln 3} - \mathrm{e}^{-2\ln 3}}{2} + 2\ln 3 - 3 \times \frac{\mathrm{e}^{\ln 3} - \mathrm{e}^{-\ln 3}}{2}$

$= \frac{1}{2}\left(\mathrm{e}^{\ln 3^2} - \mathrm{e}^{\ln 3^{-2}}\right) + \ln 9 - \frac{3}{2}\left(\mathrm{e}^{\ln 3} - \mathrm{e}^{\ln 3^{-1}}\right)$

$= \frac{1}{2}\left(\mathrm{e}^{\ln 9} - \mathrm{e}^{\ln\frac{1}{9}}\right) + \ln 9 - \frac{3}{2}\left(\mathrm{e}^{\ln 3} - \mathrm{e}^{\ln\frac{1}{3}}\right)$

$= \frac{1}{2}\left(9 - \frac{1}{9}\right) + \ln 9 - \frac{3}{2}\left(3 - \frac{1}{3}\right)$

$= \frac{40}{9} + \ln 9 - 4$

$= \frac{4}{9} + \ln 9$

25) a)

$\int \sinh^3 x \, \mathrm{d}x$

$= \int \sinh^2 x \times \sinh x \, \mathrm{d}x$

$= \int (\cosh^2 x - 1) \sinh x \, \mathrm{d}x$

$= \int (\cosh^2 x \sinh x - \sinh x) \, \mathrm{d}x$

$= \frac{1}{3}\cosh^3 x - \cosh x + c$

Alternatively:

$\sinh^3 x = \frac{1}{8}(\mathrm{e}^x - \mathrm{e}^{-x})^3$

$= \frac{1}{8}(\mathrm{e}^{3x} - 3\mathrm{e}^x + 3\mathrm{e}^{-x} - \mathrm{e}^{-3x})$

$= \frac{1}{8}(\mathrm{e}^{3x} - \mathrm{e}^{-3x}) - \frac{1}{8}(3\mathrm{e}^x - 3\mathrm{e}^{-x})$

$= \frac{1}{4}\left(\frac{\mathrm{e}^{3x} - \mathrm{e}^{-3x}}{2}\right) - \frac{3}{4}\left(\frac{\mathrm{e}^x - \mathrm{e}^{-x}}{2}\right)$

$= \frac{1}{4}\sinh 3x - \frac{3}{4}\sinh x$

So that $\int \sinh^3 x \, \mathrm{d}x$

$= \int \left(\frac{1}{4}\sinh 3x - \frac{3}{4}\sinh x\right) \mathrm{d}x$

$= \frac{1}{12}\cosh 3x - \frac{3}{4}\cosh x + c$

25) b)

$\int \cosh^3 x \, \mathrm{d}x$

$= \int \cosh^2 x \times \cosh x \, \mathrm{d}x$

$= \int (\sinh^2 x + 1) \cosh x \, \mathrm{d}x$

$= \int (\sinh^2 x \cosh x + \cosh x) \, \mathrm{d}x$

$= \frac{1}{3}\sinh^3 x + \sinh x + c$

Alternatively:

$\cosh^3 x = \frac{1}{8}(\mathrm{e}^x + \mathrm{e}^{-x})^3$

$= \frac{1}{8}(\mathrm{e}^{3x} + 3\mathrm{e}^x + 3\mathrm{e}^{-x} + \mathrm{e}^{-3x})$

$= \frac{1}{8}(\mathrm{e}^{3x} + \mathrm{e}^{-3x}) + \frac{1}{8}(3\mathrm{e}^x + 3\mathrm{e}^{-x})$

$= \frac{1}{4}\left(\frac{\mathrm{e}^{3x}+\mathrm{e}^{-3x}}{2}\right) + \frac{3}{4}\left(\frac{\mathrm{e}^x+\mathrm{e}^{-x}}{2}\right)$

$= \frac{1}{4}\cosh 3x + \frac{3}{4}\cosh x$

So that $\int \cosh^3 x \, \mathrm{d}x$

$= \int \left(\frac{1}{4}\cosh 3x + \frac{3}{4}\cosh x\right) \mathrm{d}x$

$= \frac{1}{12}\sinh 3x + \frac{3}{4}\sinh x + c$